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Course: Data Science
Exercise 1 = Select & Sorting

Question 1: Retrieving all columns

```
SELECT *
FROM employees;
```

id	First_name	Last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-04-10	Johannes
2	Brian	Lee	HR	45000	2014-03-22	Cape Tow
3	Cathy	Zulu	Finance	65000	2018-07-15	Durban
4	David	Molokeng	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesbu

Question 2: Unique departments

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

Question 3 Names ordered by salary

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

First_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Question 4 : Retrieval of Top 3 highest paid ~~First and last names ordered by salary~~

```
SELECT id,
       first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

id	First_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Question 5: Employees in the IT department.

```
SELECT id,
       First_name,
       last_name,
       department
FROM employees
WHERE department = 'IT';
```

id	First_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Question 6

```
SELECT id,
       First_name,
       last_name,
       department,
       Salary
FROM employees
WHERE department = 'Finance'
AND salary > 60 000;
```

id	First_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000 ~~salary~~ ~~65 000~~

Question 7: Employees in HR or Marketing

```
SELECT id,
       First_name,
       last_name,
       department
FROM employees
WHERE department = 'HR'
OR department = 'Marketing'
```

id	First_name	last_name	department
2	Brian	Lee	HR
4	David	Molcoena	Marketing

Question 8: Employees not in IT

```
SELECT id,
       First_name,
       last_name
       department
FROM employees
WHERE department != 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Question 9: Employees in IT, HR or Finance using IN

```
SELECT id,
       first_name,
       last_name
       department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Question 10: IT employees earning more than R65 000

```
SELECT id,
       first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg